

INPUT:

network_classification.py > ...

```
1  # Network Device and Transmission Media Classification
2
3  network_info = {
4      "Switch": {
5          "layer": "Layer 2 - Data Link",
6          "function": "Connects devices within a LAN and forwards frames using MAC addresses."
7      },
8
9      "Router": {
10         "layer": "Layer 3 - Network",
11         "function": "Connects different networks and forwards packets using IP addresses."
12     },
13
14     "Bridge": {
15         "layer": "Layer 2 - Data Link",
16         "function": "Connects and filters traffic between LAN segments using MAC addresses."
17     },
18
19     "Access Point": {
20         "layer": "Layer 2 - Data Link",
21         "function": "Provides wireless access and connects wireless devices to a wired LAN."
22     },
23
24     "Twisted Pair Cable": {
25         "layer": "Layer 1 - Physical",
26         "function": "Transmits data using electrical signals through copper wires."
27     },
28
29     "Coaxial Cable": {
30         "layer": "Layer 1 - Physical",
31         "function": "Transmits data using electrical signals through a copper conductor."
32     },
33 }
```

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network_classification.py > ...
32     },
33
34     "Fiber Optic Cable": {
35         "layer": "Layer 1 - Physical",
36         "function": "Transmits data using pulses of light through optical fibers."
37     },
38
39     "Wireless": {
40         "layer": "Layer 1 - Physical",
41         "function": "Transmits data through radio waves without physical cables."
42     }
43 }
44
45
46 def classification_report(items):
47     print("\n===== NETWORK CLASSIFICATION REPORT =====\n")
48
49     for item in items:
50
51         if item in network_info:
52             info = network_info[item]
53
54             print("Device/Medium :", item)
55             print("Layer          :", info["layer"])
56             print("Primary Role   :", info["function"])
57             print("-" * 60)
58
59         else:
60             print("Device/Medium :", item)
61             print("Status        : Unknown")
62             print("-" * 60)
63

```

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52         info = network_info[item]
53
54         print("Device/Medium :", item)
55         print("Layer          :", info["layer"])
56         print("Primary Role  :", info["function"])
57         print("-" * 60)
58
59     else:
60         print("Device/Medium :", item)
61         print("Status          : Unknown")
62         print("-" * 60)
63
64
65     items = [
66         "Switch",
67         "Router",
68         "Bridge",
69         "Access Point",
70         "Twisted Pair Cable",
71         "Coaxial Cable",
72         "Fiber Optic Cable",
73         "Wireless"
74     ]
75
76     classification_report(items)

```

OUTPUT:

===== NETWORK CLASSIFICATION REPORT =====

Device/Medium : Switch
Layer : Layer 2 - Data Link
Primary Role : Connects devices within a LAN and forwards frames using MAC addresses.

Device/Medium : Router
Layer : Layer 3 - Network
Primary Role : Connects different networks and forwards packets using IP addresses.

Device/Medium : Bridge
Layer : Layer 2 - Data Link
Primary Role : Connects and filters traffic between LAN segments using MAC addresses.

Device/Medium : Access Point
Layer : Layer 2 - Data Link
Primary Role : Provides wireless access and connects wireless devices to a wired LAN.

Device/Medium : Twisted Pair Cable
Layer : Layer 1 - Physical
Primary Role : Transmits data using electrical signals through copper wires.

Device/Medium : Coaxial Cable
Layer : Layer 1 - Physical
Primary Role : Transmits data using electrical signals through a copper conductor.

Device/Medium : Fiber Optic Cable
Layer : Layer 1 - Physical
Primary Role : Transmits data using pulses of light through optical fibers.

Device/Medium : Wireless
Layer : Layer 1 - Physical
Primary Role : Transmits data through radio waves without physical cables.
